

FACTORY DESIGN PATTERN

① Definition

② Practical

Defⁿ

Factory Pattern is used to create object without exposing the creation logic to the client.

① Simple / Practical Factory

② Factory Method

③ Abstract Factory

(most commonly used)

① Framework

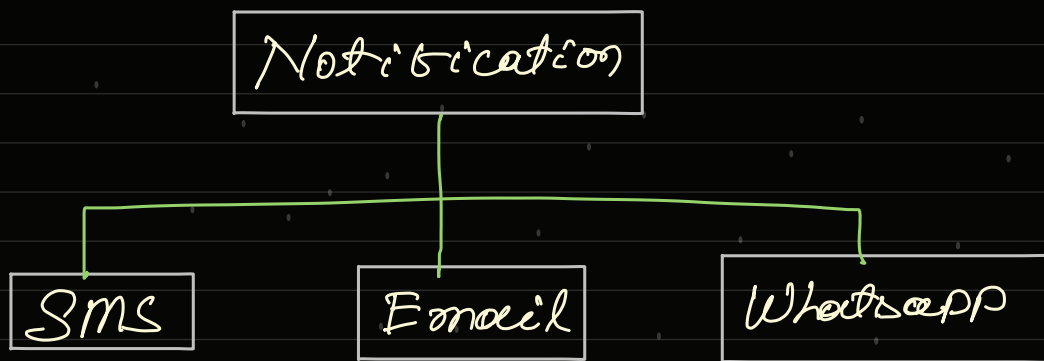
② Large application

Simple / Practical Factory Design Pattern

It allows you to create connect child class object based on some condition.

Example

Notification System



Problem Statement

- ① I have some input
- ② Based on input, I need to create child class object.
- ③ May need to create the object in multiple places in our code base.

Solution-1

String input

Notification n

if input == "sms"

n = new SMS()

if input == "email"

n = new Email()

if input == "whatsapp"

n = new WhatsApp()

Problems in solution-1

- ① Violating SRP, OCP
- ② Code repetition, because we may need to use same logic in multiple places.

Solution-2

Using Simple/Practical Factory

